

A CLASS-THREE VERSION OF THE FIFTH DIMENSION ARGUMENT

1. NOTATION

All Lie rings are Lie algebras over $\mathbb{Z}$. If L is a Lie ring, its lower central series is denoted by $(\gamma_n(L))_{n \geq 1}$. The universal enveloping algebra of L is denoted by $U(L)$, and its augmentation ideal by $\varpi(L)$.

All commutators are left-normed:

$$[x_1, x_2, \dots, x_m] = [\dots [[x_1, x_2], x_3], \dots, x_m].$$

For integers a, b , (a, b) denotes their non-negative greatest common divisor.

2. DEFINITIONS

Definition 1. The lower central series of L is defined by

$$\gamma_1(L) = L, \quad \gamma_{n+1}(L) = [L, \gamma_n(L)].$$

Definition 2. The n -th Lie dimension subring of L is

$$\delta_n(L) = L \cap \varpi(L)^n \subseteq U(L).$$

3. KNOWN FACTS

Theorem 3 (Bartholdi–Passi). *For every Lie ring L ,*

$$\delta_2(L) = \gamma_2(L), \quad \delta_3(L) = \gamma_3(L).$$

Proof. These are Theorems 4.3 and 4.5 of Bartholdi–Passi. □

4. NORMAL FORM IN NILPOTENCY CLASS THREE

Throughout the rest of the paper, let L be a finite Lie ring with $\gamma_4(L) = 0$.

Definition 4. Let A be a finite abelian group. A Smith basis of A is a finite sequence $a_1, \dots, a_m$ and integers $\varepsilon_1, \dots, \varepsilon_m$, each at least 2, such that

$$A = \bigoplus_{i=1}^m \langle a_i \rangle, \quad \langle a_i \rangle \simeq \mathbb{Z}/\varepsilon_i \mathbb{Z},$$

and

$$\varepsilon_1 \mid \varepsilon_2 \mid \dots \mid \varepsilon_m,$$

with trivial invariant factors 1 not listed. The integers ε_i are called elementary divisors.

Choose elements $x_i \in L$ whose images form a Smith basis of

$$L/\gamma_2(L) = \gamma_1(L)/\gamma_2(L),$$

and let d_i be the corresponding elementary divisors. Let F be the free Lie ring on lifts of the x_i , and let $F \rightarrow L$ be the induced epimorphism. Put $R = \ker(F \rightarrow L)$. Since $\gamma_4(L) = 0$,

$$(1) \quad \gamma_4(F) \subseteq R, \quad L \simeq F/R.$$

Choose a basis of $\gamma_2(F)/\gamma_3(F)$ consisting of elements y_k and supplementary elements which map to zero in $\gamma_2(L)/\gamma_3(L)$, such that the kernel of

$$\gamma_2(F)/\gamma_3(F) \longrightarrow \gamma_2(L)/\gamma_3(L)$$

is generated by the non-zero multiples $e_k y_k$ and by the supplementary elements. The images of the y_k 's form a Smith basis of $\gamma_2(L)/\gamma_3(L)$, with elementary divisors e_k .

Similarly, choose a basis of $\gamma_3(F)/\gamma_4(F)$ consisting of elements z_l and supplementary elements which map to zero in $\gamma_3(L)$, such that the kernel of

$$\gamma_3(F)/\gamma_4(F) \longrightarrow \gamma_3(L)$$

is generated by the non-zero multiples $f_l z_l$ and by the supplementary elements. The images of the z_l 's form a Smith basis of $\gamma_3(L)$, with elementary divisors f_l .

Thus, in L ,

$$(2) \quad \begin{aligned} d_i x_i &= B_i + C_i, & B_i &= \sum_k b_{ik} y_k, & C_i &= \sum_l c_{il} z_l, \\ e_k y_k &= \sum_l \beta_{kl} z_l, & f_l z_l &= 0. \end{aligned}$$

In F , put

$$(3) \quad r_i = d_i x_i - B_i - C_i, \quad s_k = e_k y_k - \sum_l \beta_{kl} z_l, \quad t_l = f_l z_l.$$

Then $r_i, s_k, t_l \in R$.

Let

$$\mathfrak{r} = U(F)RU(F).$$

The natural map $U(F) \rightarrow U(L)$ induces

$$(4) \quad U(F)/\mathfrak{r} \simeq U(L).$$

Since F is free on the x_i , its enveloping algebra is the free associative algebra $\mathbb{Z}\langle x_i \rangle$. We use its word-length filtration. Thus $\varpi(F)^n$ consists of sums of words of length at least n , and $\gamma_n(F) \subseteq \varpi(F)^n$.

Theorem 5 (PBW basis modulo $\varpi(F)^m$). *Let F be a free Lie ring and $m \geq 2$. For each $1 \leq n < m$, choose a $\mathbb{Z}$ -basis of $\gamma_n(F)/\gamma_{n+1}(F)$ and representatives in $\gamma_n(F)$. After ordering the union of these representatives, the ordered products*

$$b_1 \cdots b_s, \quad b_1 \leq \cdots \leq b_s, \quad \ell(b_1) + \cdots + \ell(b_s) < m,$$

including 1, form a $\mathbb{Z}$ -basis of $U(F)/\varpi(F)^m$. Here $\ell(b) = n$ for $b \in \gamma_n(F)$ representing a basis element of $\gamma_n(F)/\gamma_{n+1}(F)$.

Proof. This is the filtered PBW theorem for $\text{gr}_\varpi U(F) \simeq U(\text{gr}_\gamma F)$. □

Construction 6 (PBW basis modulo $\varpi(F)^5$). Use the x_i 's in degree 1, the y_k 's followed by the supplementary degree 2 elements in degree 2, the z_l 's followed by the supplementary degree 3 elements in degree 3, and complete this to a PBW basis modulo $\varpi(F)^5$ arbitrarily in degree 4. Order the named elements so that

$$x_1 < x_2 < \cdots < y_1 < y_2 < \cdots < z_1 < z_2 < \cdots,$$

with supplementary degree 2 elements placed after the y_k 's and before the z_l 's, and supplementary degree 3 elements placed after the z_l 's.

The Smith choices and (1) imply that, modulo $\gamma_5(F)$, every element of R is a sum of elements

$$(5) \quad r_i, \quad s_k, \quad t_l,$$

together with relation representatives whose first non-zero component is supplementary in degrees 2 or 3, and an element of $\gamma_4(F)$.

Let $\langle K \rangle_{\mathbb{Z}}$ be the $\mathbb{Z}$ -subgroup of $U(F)$ generated by

$$(K) \quad r_i x_j y_k \ (i \leq j), \quad x_i x_j s_k \ (i \leq j), \quad r_i z_l, \quad x_i t_l, \quad s_k y_k, \quad s_k y_l, \quad y_k s_l \ (k < l).$$

Every generator in (K) lies in $\mathfrak{r} \cap \varpi(F)^4$.

Definition 7. The following elements of $U(F)$, taken modulo $\varpi(F)^5$, are used:

- (I) t_l ,
- (II) $\frac{d_j}{d_i} r_i x_j - x_i r_j, \quad i < j$,
- (III) $r_i y_k$,
- (IV) $x_i s_k$,
- (V) $r_i x_j x_k, \quad i \leq j \leq k$,
- (VI) $x_i r_j x_k, \quad i < j \leq k$,
- (VII) $x_i x_j r_k, \quad i \leq j < k$.

Lemma 8 (Reduction of relation-products). *Let*

$$f \in F \cap \varpi(F)^3 = \gamma_3(F)$$

and let $\theta \in \mathfrak{r}$ satisfy

$$\theta - f \in \varpi(F)^5.$$

Then there are $\eta \in R$ and a $\mathbb{Z}$ -linear combination Θ of the elements (I)–(VII) such that

$$(6) \quad \Theta - (f - \eta) \in \varpi(F)^5 + \langle K \rangle_{\mathbb{Z}}.$$

Proof. The equality $F \cap \varpi(F)^3 = \gamma_3(F)$ follows from Theorem 3. By (5), every element of $\mathfrak{r}$ is, modulo $\varpi(F)^5$, a sum of products aqb , where q is one of the listed relation representatives and a, b are ordered PBW products from Construction 6. A factor from $\gamma_4(F)$ with a nonconstant PBW factor lies in $\varpi(F)^5$; without such a factor it is absorbed into R .

Terms containing supplementary relation factors or supplementary PBW factors have, below $\varpi(F)^5$, ordered products with a supplementary factor. Their coefficients vanish in

$$\frac{\varpi(F)^3}{(F \cap \varpi(F)^3) + \varpi(F)^5},$$

because $\theta - f \in \varpi(F)^5$ and $f \in F$. Commutators produced while ordering either lie in R , lie in $\varpi(F)^5$, or give one of the generators in (K) . The only exceptional ordering $[y_{\text{sup}}, x_i]$ has zero image in $\gamma_3(L)$, hence is a sum of $f_l z_l$ -parts and supplementary degree 3 parts modulo $R + \gamma_4(F)$, and is handled in the same way.

The remaining relation factors are r_i, s_k, t_l . Moving the relation factor by $uq = qu + [u, q]$, $[u, q] \in R$, and ordering the other PBW factors leaves, modulo $R + \varpi(F)^5 + \langle K \rangle_{\mathbb{Z}}$, only the seven types listed in Definition 7. The four- x products and the products $r_i x_j, x_i r_j$ are reduced by the adjacent syzygies $(d_j/d_i, -1)$, which are exactly type (II). This proves (6). $\square$

Corollary 9 (Kernel reduction). *Let $f \in F$ represent an element of $\delta_5(L)$ in F/R . Then, after replacing f by another lift modulo R , there is a $\mathbb{Z}$ -linear combination Θ of (I)–(VII) such that*

$$(7) \quad \Theta - f \in \varpi(F)^5 + \langle K \rangle_{\mathbb{Z}}.$$

Proof. The image of f in L lies in $\delta_5(L) \subseteq \delta_3(L) = \gamma_3(L)$. Since $F \rightarrow L$ is surjective, the induced map $\gamma_3(F) \rightarrow \gamma_3(L)$ is surjective, so we may assume $f \in \gamma_3(F) = F \cap \varpi(F)^3$. The image of f in $U(L)/\varpi(L)^5$ is zero, hence there is $\theta \in \mathfrak{r}$ with $\theta - f \in \varpi(F)^5$. Apply Lemma 8 and absorb the resulting $\eta \in R$ into the choice of f . $\square$

Proposition 10 (Coefficient comparison). *Let*

$$u_{ij}, v_{ik}, v'_{ik}, w_{ijk}, w'_{ijk}, w''_{ijk} \in \mathbb{Z}$$

with u_{ij} defined for $i < j$, w_{ijk} for $i \leq j \leq k$, w'_{ijk} for $i < j \leq k$, and w''_{ijk} for $i \leq j < k$. Let Θ be the corresponding linear combination of (II)–(VII), together with an arbitrary linear combination of (I). If there exists $f \in F$ such that

$$(8) \quad \Theta - f \in \varpi(F)^5 + \langle K \rangle_{\mathbb{Z}},$$

then

$$(B) \quad \sum_{i < h} u_{ih} b_{hk} - \sum_{h < i} u_{hi} \frac{d_i}{d_h} b_{hk} + d_i v_{ik} + e_k v'_{ik} = 0$$

for all i, k . Moreover the image of f in $L = F/R$ is the negative of

$$(9) \quad \sum_{i < j} u_{ij} d_j [x_i, x_j].$$

Proof. Expand Θ in $\varpi(F)^3/\varpi(F)^5$ using the PBW basis from Construction 6. Since $f \in F$, the coefficient of the ordered product $x_i y_k$ in Θ is zero. The subgroup $\langle K \rangle_{\mathbb{Z}}$ does not change this coefficient. The coefficient of $x_i y_k$ is exactly

$$\sum_{i < h} u_{ih} b_{hk} - \sum_{h < i} u_{hi} \frac{d_i}{d_h} b_{hk} + d_i v_{ik} + e_k v'_{ik},$$

which gives (B).

It remains to record the part of Θ which lies in F after passage to F/R . For type (II),

$$\frac{d_j}{d_i} r_i x_j - x_i r_j \equiv -d_j [x_i, x_j] \pmod{R}.$$

For type (III), the only term in F is $-[B_i, y_k]$, which is zero in L because $[\gamma_2(L), \gamma_2(L)] \subseteq \gamma_4(L) = 0$. For type (V), the only term in F is $-[B_i, x_j, x_k]$, also zero in L . Types (I), (IV), (VI), and (VII) contribute nothing to F/R . This proves (9). $\square$

Theorem 11 (Normal form). *Every $a \in \delta_5(L)$ can be written in the form*

$$(10) \quad a = \sum_{i < j} u_{ij} d_j [x_i, x_j],$$

where the integers u_{ij} satisfy (B) for some integers v_{ik}, v'_{ik} .

Proof. Choose a lift $f \in F$ of a . By Corollary 9, after replacing f modulo R there is a linear combination Θ as in Proposition 10 satisfying (8). The proposition gives (B) and shows that a is the negative of (9). Replace all coefficients in Θ by their negatives. $\square$

5. THE EXPONENT IN NILPOTENCY CLASS THREE

Theorem 12. *Let L be a finite Lie ring with $\gamma_4(L) = 0$. Then*

$$2\delta_5(L) = 0.$$

Proof. Let $a \in \delta_5(L)$. Write a as in Theorem 11:

$$a = U = \sum_{i < j} u_{ij} d_j [x_i, x_j],$$

with (B). Since $d_j x_j = B_j + C_j$ in L , and $[x_i, C_j] \in \gamma_4(L) = 0$,

$$(11) \quad U = \sum_{i < j} u_{ij} [x_i, B_j] = \sum_{i, k} \left(\sum_{i < j} u_{ij} b_{jk} \right) [x_i, y_k].$$

Using (B), the coefficient in parentheses is

$$\sum_{h < i} u_{hi} \frac{d_i}{d_h} b_{hk} - d_i v_{ik} - e_k v'_{ik}.$$

The last two terms give no contribution, since

$$d_i [x_i, y_k] = [d_i x_i, y_k] = [B_i + C_i, y_k] = 0,$$

and

$$e_k [x_i, y_k] = [x_i, e_k y_k] = \left[x_i, \sum_l \beta_{kl} z_l \right] = 0.$$

Therefore

$$\begin{aligned} U &= \sum_{h < i, k} u_{hi} \frac{d_i}{d_h} b_{hk} [x_i, y_k] \\ &= \sum_{h < i} u_{hi} \frac{d_i}{d_h} [x_i, B_h] = \sum_{h < i} u_{hi} d_i [x_i, x_h] = -U. \end{aligned}$$

Thus $2U = 0$, and hence $2a = 0$. □

Corollary 13. *For every finite Lie ring L with $\gamma_4(L) = 0$,*

$$6\delta_5(L) = 0.$$

REFERENCES

- [1] L. Bartholdi and I. B. S. Passi, *Lie dimension subrings*, J. Algebra Appl. 14 (2015), no. 6, 1550042. Also available as arXiv:1308.2118.
- [2] K.-I. Tahara, *The augmentation quotients of group rings and the fifth dimension subgroups*, J. Algebra 71 (1981), 141–173.